

SHASHANK SINGH

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LinkedIn

GitHub

Portfolio

Professional Summary

AI/ML Engineer with 2 years of experience developing **production-oriented Generative AI, Agentic AI, and Retrieval-Augmented Generation (RAG)** applications. Experienced in **LLM APIs, semantic retrieval, GraphRAG, vector databases, AI copilots, FastAPI microservices, MCP orchestration, and AI evaluation**, with strong skills in Python, LangChain, embeddings, prompt engineering, API integration, data processing, validation, observability, and scalable enterprise AI workflows.

Technical Skills

Languages: Python, Java, SQL

AI & GenAI: LLMs, Generative AI, Agentic AI, RAG, GraphRAG, Multi-Agent Systems, Prompt Engineering, Semantic Retrieval, Hybrid Search, Embeddings, NLP, Transformers, Confidence Scoring, Retrieval Evaluation, AI Validation, Hallucination Mitigation

Frameworks & Libraries: LangChain, LangGraph, FastAPI, Flask, PyTorch, Hugging Face, Scikit-Learn, Pydantic

Conversational & Voice AI: Rasa, Chatbots, Voice AI Workflows, Intent Detection, Entity Extraction, Dialogue Management, Multi-Turn Conversations, Tool Calling, Function Calling, Transcript Processing, TTS/SSML Concepts

Backend & Integration: FastAPI, Java, Spring Boot, REST APIs, Microservices, MCP, SDK Development, API Integration, Workflow Automation

Data & Vector Engineering: MySQL, MongoDB, FAISS, ChromaDB, Pinecone, Vector Search, Data Ingestion, ETL/ELT, Data Transformation, Structured & Unstructured Data

Cloud, DevOps & Engineering: AWS, Azure, Docker, Git, GitHub, CI/CD, Postman, Logging, Observability, Testing, Error Handling

Experience

Tata Consultancy Services Ltd.

AI/ML Developer / Product Engineer

Oct 2024 – Present

Gurugram, Haryana

- Built production-oriented **Generative AI applications** using LLM APIs, RAG, semantic retrieval, embeddings, and vector search to deliver knowledge-grounded enterprise workflows.
- Orchestrated **AI-agent workflows, tool calling, retrieval, validation, and multi-step reasoning** using LangChain and MCP for reliable interaction with enterprise tools and services.
- Integrated **REST APIs, Java services, Python SDKs, and enterprise backend systems** to enable AI agents to invoke tools and automate business workflows.
- Developed **document ingestion, chunking, embedding generation, vector indexing, and ETL-style pipelines** across structured and unstructured enterprise data.
- Built **FastAPI microservices and backend integrations** with API validation, error handling, logging, observability, and reliability workflows for enterprise AI deployments.
- Applied **prompt engineering, response validation, evaluation, and confidence-based checks** to improve LLM output quality, reliability, and production readiness.

Soven Developer

Software Developer

Jun 2024 – Sept 2024

Remote

- Developed responsive application interfaces and integrated REST APIs, applying Python/JavaScript fundamentals and basic data-processing concepts to support reliable user-facing workflows.
- Collaborated with cross-functional teams to troubleshoot application issues, improve frontend-backend integration, and support maintainable application development.

Projects

Enterprise AI SQL Copilot

Python, FastAPI, Agentic AI, RAG, GraphRAG, FAISS

- Built an enterprise **Text-to-SQL AI Copilot** combining Agentic AI, RAG, GraphRAG, semantic retrieval, and schema-aware SQL generation for complex analytical queries.
- Orchestrated agents for **intent detection, schema linking, query planning, SQL generation, validation, error correction, and confidence scoring** to improve query reliability.
- Developed ingestion, embedding, vector indexing, GraphRAG context expansion, retrieval evaluation, and validation pipelines using **FAISS, ChromaDB, and Pinecone**.
- Created reliability workflows for SQL correctness, contextual relevance, explainability, and retrieval quality evaluation.

Enterprise AI Conversational Platform

Rasa, LangChain, FastAPI, FAISS, MCP

- Developed an enterprise **AI assistant platform** using LangChain, FastAPI, LLM APIs, RAG, and retrieval pipelines for knowledge-grounded workflow automation.
- Implemented **semantic retrieval, context assembly, validation, and multi-step workflow orchestration** to improve response relevance, consistency, and reliability.
- Connected enterprise APIs and custom Python SDKs with **MCP-based tool orchestration** to enable AI-agent actions and backend workflow automation.

Certifications & Training

- Claude Certified Architect – Professional** – Anthropic
- AWS Certified AI Practitioner** – Amazon Web Services
- Martian Summer Internship Training** – Persistent Systems | DBMS, Linux/Unix, Data Structures & Algorithms, Python, Java

Education

Siksha 'O' Anusandhan University

B.Tech in Computer Science & Engineering | CGPA: 8.50/10

Aug 2020 – Jul 2024

Bhubaneswar, Odisha